Roll No.: BL.AI.U4AD24140

Amrita Vishwa Vidyapeetham

Amrita School of Artificial Intelligence, Faridabad

B.Tech Mid-Term Examinations – March 2025

Second Semester

Artificial Intelligence & Data Science

23ECE113 Introduction to Electronics

Duration: Two hours Maximum: 50 Marks

Course Outcomes (COs):

CO	Course Outcomes
CO01	Explain the basic concepts of analogy and digital electronics.
CO02	Design and implement various electronic circuits using diodes and transistors and demonstrate an understanding of their applications in electronic devices and systems.
CO03	Design and implement operational amplifier circuits for a range of applications and demonstrate an understanding of their operation and behavior in electronic systems.
CO04	Model and analyze engineering problems and demonstrate the ability to propose and implement electronic solutions to these problems.
CO05	To integrate theoretical knowledge with practical applications through an end-term project-based learning approach, where students design, prototype, and test electronic circuits for AI and data science-related applications.

Answer all questions

1. A 12V unregulated power supply is connected to a Zener diode voltage regulator. The Zener breakdown voltage is 6V, and the series resistance is 100Ω. Calculate the total supplied current, current across series resistance, and Zener current if the load resistance is 500Ω. Also, discuss the significance of the Zener diode as a voltage regulator. [8 marks] [CO04 BTL 3]

2. Explain how doping improves the conductivity of a semiconductor. [2 marks] [CO01 BTL 1]

3. Design a biased clamping circuit to shift a 5V peak sine wave by +3V. Explain the circuit operation. [8 marks] [CO02 BTL 6]

4. Define electrical conduction in semiconductors. How does it differ from conduction in insulators? [2 marks] [CO01 BTL 1]

5. Explain the forward and reverse biasing using the IV characteristic curve in a silicon-based diode. [5 marks] [CO01 BTL 2]

6. What is the role of Bipolar Junction Transistors in electronic circuits? [5 marks] [CO02 BTL 4]

7. Write pseudocode in Python programming language to generate combinational clipper circuit output. Draw the required circuit diagram to generate its output. [8 marks] [CO04 BTL 6]

8. Discuss the role of avalanche breakdown in diodes. [2 marks] [CO01 BTL 2]

9. In a circuit simulation software (such as TinkerCad/Multism), how would you set up and analyze the waveform of a full-wave rectifier? List the steps and expected output for a full-wave rectifier operating with a 50 Hz ±10 V AC supply. How does a capacitor filter improve the rectified output? [8 marks] [CO05 BTL 6]

10. In a practical clamping circuit, why is a resistor placed parallel to the diode? What role does it play in maintaining proper clamping? [2 marks] [CO04 BTL 5]

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CO	Marks	BTL	Marks
CO01	**11**	BTL 1	**4**
CO02	**13**	BTL 2	**7**
CO03	**-**	BTL 3	**8**
CO04	**18**	BTL 4	**5**
CO05	**08**	BTL 5	**2**
CO06	**-**	BTL 6	**24**

Bloom's Taxonomy Levels (attached for reference)

Level 1 – Remember

Level 2 - Understand

Level 3 – Apply

Level 4 - Analyze

Level 5 - Evaluate

Level 6- Create